

Roll No.

TAS-708

**B. TECH. (SEVENTH SEMESTER)
MID SEMESTER EXAMINATION, Oct., 2022**

FLIGHT DYNAMICS

Time : 1½ Hours

Maximum Marks : 50

Note : (i) Answer all the questions by choosing any *one* of the sub-questions.
(ii) Each question carries 10 marks.

1. (a) Explain the difference between streamline and bluff body.

10 Marks (CO1)

OR

(b) What is the difference between skin friction and pressure drag ?

10 Marks (CO1)

2. (a) What do you understand by standard atmosphere ? Why is it used ?

10 Marks (CO1)

OR

(b) With the help of a suitable diagram, explain the layers of standard atmosphere.

10 Marks (CO1)

3. (a) What type of propulsion is suitable for a slow moving aircraft ? Why ?

10 Marks (CO1 & CO2)

P. T. O.

(2)

OR

(b) Why is a propeller more efficient than a jet engine ?

10 Marks (CO1 & CO2)

4. (a) How can the thrust requirement for a steady level flight can be estimated ?

10 Marks (CO2)

OR

(b) What is the drag polar and induced drag for an aircraft ?

10 Marks (CO2)

5. (a) Draw a diagram for the longitudinal dynamics of an aircraft and label all the forces and relevant angles.

10 Marks (CO2)

OR

(b) Derive the governing differential equations of an aircraft for the longitudinal dynamics.

10 Marks (CO2)